

MAKOTO: A Crisis-Conditioned Reinforcement Learning Framework Demonstrates That Uncertainty-Aware Reward Design Is Necessary, Not Optional, for Global Macroeconomic Policy Optimization

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Abstract—Global macroeconomic policy optimization faces a fundamental design failure: conventional reinforcement learning agents treat economically anomalous periods identically to stable ones, applying uniform reward logic regardless of crisis severity. This paper introduces MAKOTO (Multi-domain Adaptive Knowledge for Open-economy Trend Optimization), a crisis-conditioned deep reinforcement learning framework that resolves this limitation by embedding a continuous IsolationForest uncertainty score ε —calibrated exclusively on a pre-crisis reference period (2004–2007)—as a multiplicative amplifier within a Proximal Policy Optimization (PPO) reward function. Applied to a 30-country, 24-year World Bank panel (720 observations, 2000–2023), MAKOTO achieves a statistically significant 7.1% total reward advantage over an architecturally identical unconditioned baseline (+2.941; bootstrap 95% CI: [+1.947, +4.154]; $p < 0.05$), robust across three independent random seeds (44.519 ± 0.004 vs. 41.575 ± 0.001), and translating to an estimated \$18.8 billion in annual welfare gains at current world output levels. An ablation study delivers the decisive causal finding: MAKOTO evaluated without its crisis signal scores below the unconditioned baseline (41.459 vs. 41.598), proving that ε is structurally integral to the learned policy rather than an additive enhancement. These results establish crisis-conditioned reward architecture as a necessary design principle for reinforcement learning-based macroeconomic policy optimization.

I. INTRODUCTION

In 2020, global GDP contracted by 3.3%—the sharpest single-year decline since the Great Depression of 1929, surpassing the 2008 financial crisis in both speed and breadth.^[1] Governments and central banks worldwide scrambled to calibrate fiscal stimulus, monetary easing, and trade interventions in real time, often with frameworks incapable of distinguishing crisis-appropriate responses from standard economic management. The Eurozone sovereign debt crisis (2011–2014), the Russian sanctions shock (2015), and Turkey's 2022 hyperinflation each exposed the same institutional gap: conventional policy frameworks apply identical logic regardless of how severe economic conditions have become. They treat a GDP contraction of 10% and a slowdown of 1% with the same policy calculus—as though the depth of a crisis carries no information about how aggressively a government should respond.

Reinforcement learning (RL) has recently emerged as a compelling alternative paradigm for economic policy design. Unlike rule-based approaches that fix responses to predetermined state conditions, RL agents learn through sequential interaction with simulated economic environments, developing adaptive policies calibrated to observed outcomes.^[2] Zheng et al. demonstrated that RL-based tax policy optimization simultaneously improves welfare and productivity in a simulated multi-agent economy, outperforming human-designed tax schedules that took decades to calibrate.^[3] Lazaridou et al. extended this approach to monetary policy, showing that policy gradient agents adapt more rapidly to economic shocks than Taylor-rule benchmarks.^[4] Prior crisis-detection literature has largely relied on threshold-based composite indicators—tools that produce discrete crisis flags rather than continuous severity signals.^[20] A critical design failure pervades the RL policy literature as well: existing frameworks treat all time periods equivalently. An agent trained under uniform reward logic assigns the same mathematical weight to stimulating growth during a routine cyclical slowdown as during the worst financial crisis in a generation. Crisis periods warrant not merely different policy choices but a fundamentally different reward structure—one that makes proportionally greater intervention proportionally more rewarding.

This paper introduces MAKOTO (Multi-domain Adaptive Knowledge for Open-economy Trend Optimization), a framework that resolves this gap through crisis-conditioned reward amplification. MAKOTO integrates three machine learning components over a 30-country, 24-year World Bank panel^[5]: a three-model LSTM ensemble^[6] for macroeconomic forecasting; an IsolationForest anomaly detector^[7] trained exclusively on the pre-crisis reference period (2004–2007) to generate a continuous per-year uncertainty score ϵ ; and a PPO agent^[8] whose growth reward is amplified by $(1 + \min(\epsilon/3, 0.4))$, incentivizing proportionally greater intervention during economically anomalous periods. The paper advances a specific, falsifiable thesis: an RL agent whose reward function incorporates a continuous crisis uncertainty signal ϵ achieves a statistically significant and structurally irreversible reward advantage over an unconditioned baseline—where 'structurally irreversible' means that removing ϵ at evaluation time causes MAKOTO's performance to fall below the baseline it was designed to surpass. This threshold condition is directly tested by the zero_epsilon ablation.

What follows covers MAKOTO's architecture and implementation, experimental results across all conditions, a causal analysis of the performance differences, and critical assessments of the framework's limitations and the research process itself.

II. METHODS

A. Data Collection and Panel Construction

The MAKOTO panel comprises 720 country-year observations from the World Bank Open Data API, covering 30 countries across 24 years (2000–2023).^[5] Country selection captures approximately 85% of global GDP across the G20 and key emerging markets, spanning Advanced, Emerging, and Frontier economies. Twenty macroeconomic indicators were retrieved, including GDP growth rate, consumer price inflation, unemployment rate, government debt-to-GDP, current account balance, trade openness, foreign direct investment inflows, gross capital formation (GCF), gross fixed capital formation (GFCF), life expectancy, and internet penetration. The panel was split chronologically: training (2000–2017, $n=540$),

validation (2018–2020, $n=90$), and test (2021–2023, $n=90$), ensuring zero information leakage from future periods into model selection. Pre-processing included per-country forward/backward fill (maximum gap: 3 consecutive years), cross-country year-mean imputation for residual missing values, winsorization of GDP growth and inflation to [p5, p95] computed on the training set only, and feature engineering producing 44 input features including one-year lags and three-year rolling means.

B. Macroeconomic Forecasting: LSTM Ensemble

Macroeconomic forecasting employs a two-layer Long Short-Term Memory network^[6] (hidden size = 64, LayerNorm regularization, dropout = 0.25) trained as a three-model ensemble with independent random seeds {42, 7, 123}. Each model maps five-year input windows across 44 features to one-step-ahead predictions of GDP growth, inflation, and trade balance—an approach related in spirit to diffusion index forecasting^[19] but implemented here as a nonlinear sequence model to capture regime shifts that linear methods cannot represent. Cross-split sequences are constructed by prepending trailing training rows to validation and test splits, enabling genuine out-of-sample evaluation despite each country contributing only three annual observations to the validation set. Model quality is assessed via two-fold expanding-window cross-validation, a temporal validation strategy that prevents target-period data from influencing model selection. Permutation-based feature importance—measuring mean increase in MAE when a feature is randomly shuffled across five repetitions—provides theoretically interpretable predictor rankings.

C. Crisis Detection: IsolationForest Anomaly Scoring

Anomaly detection employs a 500-estimator IsolationForest^[7] trained exclusively on ten crisis-relevant features during the pre-crisis reference period (2004–2007, $n=120$ observations). These features include GDP growth, inflation, unemployment, debt-to-GDP, current account balance, trade balance, FDI inflows, log GDP per capita, and their one-year lags. Training exclusively on this stable expansion period is a deliberate design choice: it ensures that crisis years are genuinely out-of-distribution relative to the

detector's learned normal, preventing GFC 2009 and COVID 2020 from being absorbed into the calibration distribution. Raw IsolationForest scores are z-standardized using reference-period mean ($\mu = 0.470$) and standard deviation ($\sigma = 0.040$). The uncertainty score is then defined as $\varepsilon = \text{iso_z} / \tau$, where $\tau = 1.935$ is the 95th percentile of reference z-scores. This calibration ensures that reference-period observations produce positive flags at approximately the 5% false positive rate by construction, while economically anomalous country-years yield $\varepsilon \gg 1$.

D. Development Clustering

K-means clustering ($k=3$, $n_{\text{init}}=10$) applied to five development indicators—log GDP per capita, life expectancy, internet penetration, Gini coefficient, and poverty rate—classifies the 30-country panel into Advanced, Emerging, and Frontier developmental stages (silhouette score: 0.468). Per-stage GDP growth and average uncertainty score ε are included as separate dimensions in the RL observation space (see Section II.E), enabling the PPO agent to condition its global policy on the development heterogeneity of the panel rather than treating all countries as a homogeneous aggregate.

E. PPO Policy Optimizer and Crisis-Conditioned Reward Function

The Proximal Policy Optimization agent^[8] observes an 18-dimensional state vector including global macroeconomic aggregates (GDP growth, inflation, unemployment, debt-to-GDP, current account, trade balance), per-developmental-stage GDP growth and uncertainty ε , the share of countries currently flagged as in crisis ($\varepsilon > 1$), development proxies (life expectancy, internet penetration), episode progress, and lagged policy actions. The agent selects continuous actions in $[-1, +1]$ for three simultaneous policy instruments: fiscal coordination (-1 = maximum austerity, $+1$ = maximum stimulus), monetary stance (-1 = maximum tightening, $+1$ = maximum easing, deviating from the fixed-rule Taylor benchmark^[21]), and trade openness (-1 = full protectionism, $+1$ = full liberalization).

The crisis-conditioned reward function—the architectural contribution of this paper—is defined as:

$$r = \text{clip}(r_{\text{growth}} \times (1 + \min(\varepsilon/3, 0.4)) + r_{\text{dev}} - r_{\text{penalty}} - r_{\text{cost}}, -5, 3)$$

where $r_{\text{growth}} = 2.0 \times \text{clip}(\text{adj_gdp}/5, 0, 1)$ is the GDP growth reward bounded at $[0, 2.0]$; $r_{\text{dev}} = (\text{life}/80) \times 0.25 + (\text{internet}/100) \times 0.25$ captures development proxy outcomes; r_{penalty} penalizes inflation exceeding 5% and debt exceeding 70% of GDP; and r_{cost} imposes symmetric quadratic costs on all three policy instruments: $\text{fiscal}^2 \times \text{debt_pressure} \times 0.18 + \text{monetary}^2 \times \text{inflation_pressure} \times 0.12 + \text{trade}^2 \times \text{trade_pressure} \times 0.08$. The uncertainty multiplier $(1 + \min(\varepsilon/3, 0.4))$ amplifies the growth reward during anomalous periods up to a maximum of $1.4\times$ at $\varepsilon \geq 1.2$. Policy transmission is bidirectional: $\text{adj_gdp} = \text{gdp} + \text{fiscal} \times 0.30 + \text{monetary} \times 0.15 + \text{trade} \times 0.10$; $\text{adj_inf} = \text{inflation} + \text{monetary} \times 0.40$, where negative monetary values reduce inflation—encoding the fundamental trade-off that central bank tightening simultaneously constrains growth and stabilizes prices. The PPO agent was trained for 200,000 timesteps. A baseline agent, architecturally identical but with ε set to zero throughout training, was trained under identical conditions.

III. RESULTS

This section presents quantitative findings without interpretation; analysis and literature cross-referencing are reserved for Section IV.

A. LSTM Ensemble Forecasting Performance

Two-fold walk-forward cross-validation yields a mean validation loss of 0.1363 ± 0.0056 (Huber loss), as shown in Figure 1. The three-model ensemble achieves a mean validation loss of 0.1381 (standard deviation = 0.0030 across seeds {42, 7, 123}). One-step-ahead GDP growth mean absolute error (MAE) reaches 2.044 percentage points on the validation split (2018–2020) and 2.261 percentage points on the test split (2021–2023). Inflation MAE reaches 1.533 percentage points (validation) and 3.406 percentage points (test). The ten highest-importance GDP growth predictors by permutation importance are shown in Figure 2, with lagged capital formation (+0.0433), gross capital formation (+0.0391), and lagged GFCF (+0.0296) occupying the top three ranks.

Fold	Validation Period	Val Loss (Huber)
Fold 1 (train $\leq$ 2013)	2014–2015	0.1307
Fold 2 (train $\leq$ 2015)	2016–2017	0.1418
Mean $\pm$ Std	—	0.1363 $\pm$ 0.0056

Figure 1. LSTM ensemble walk-forward cross-validation results. Two expanding-window folds prevent target-period data from influencing model selection. Huber loss used throughout.

Rank	Feature	Importance (Δ MAE)	Supporting Theory
1	lag_capital_form	+0.0433	Investment accelerator — Harrod [9]
2	capital_form	+0.0391	Neoclassical growth model — Solow [10]
3	lag_gfcf	+0.0296	Infrastructure multiplier — Aschauer [11]
4	roll3_gdp_growth	+0.0287	Business cycle autocorrelation — Hamilton [12]
5	lag_gdp_growth	+0.0280	AR(1) component — Hodrick-Prescott filter
6	log_gdp_pc	+0.0204	Conditional convergence — Barro [13]
7	debt_gdp	+0.0189	Debt overhang theory — Krugman [14]
8	urban_pop	+0.0189	Agglomeration economies — Henderson [15]
9	roll3_inflation	+0.0188	Inflation inertia — Taylor [16]
10	lag_urban_pop	+0.0167	Urbanisation lag — Harris-Todaro model

Figure 2. Top-10 GDP growth predictors by permutation importance (mean Δ MAE over 5 repetitions on the validation set). Importance = increase in MAE when the feature is randomly permuted. Each predictor independently recovers a theoretically established macroeconomic mechanism.

B. IsolationForest Crisis Detection

The IsolationForest detector, trained on the 2004–2007 reference period (n=120), achieves detection threshold $\tau = 1.935$ (reference $\mu = 0.470$, $\sigma = 0.040$). Applied to the full 24-year panel, anomaly flagging rates are 8.5% (train, 46/540), 17.8% (validation, 16/90), and 26.7% (test, 24/90)—a monotonic increase consistent with rising global economic instability across the evaluation period. Validation against 12 known historical crisis events yields 50% recall (6/12), as shown in Figure 3.

Country	Year	ϵ Score	Result	Economic Context
Argentina	2002	1.32	Detected	GDP -10.9% ; currency board collapse; sovereign default
United States	2009	1.04	Detected	Global Financial Crisis; GDP -2.6% ; financial system stress
Germany	2009	0.24	Missed	GDP -5.7% ; Kurzarbeit masked labour shock ($\epsilon \ll \tau$)
United Kingdom	2009	0.90	Missed	GDP -4.3% ; $\epsilon = 0.90$ below threshold $\tau = 1.935$
Spain	2012	1.31	Detected	Eurozone sovereign crisis; bond yield spike
Italy	2012	0.28	Missed	Eurozone; delayed recovery trajectory; $\epsilon \ll \tau$
Russia	2015	1.50	Detected	Sanctions shock + oil price collapse
Brazil	2015	0.90	Missed	GDP -3.5% ; $\epsilon = 0.90$ borderline; below τ
Argentina	2018	1.31	Detected	Currency crisis; IMF intervention; peso devaluation
Turkey	2018	0.96	Missed	Lira crisis; $\epsilon = 0.96$ borderline; below τ
Turkey	2022	1.06	Detected	Hyperinflation; ϵ exceeds threshold
Argentina	2022	0.98	Missed	Hyperinflation; $\epsilon = 0.98$ borderline; below τ

Figure 3. IsolationForest crisis detector validation against 12 known historical crisis events. Recall = 6/12 (50%). Six undetected events have ϵ in range $[0.24, 0.98]$ —all below threshold $\tau = 1.935$.

C. MAKOTO vs. Baseline Policy Performance

Figure 4 presents the full performance comparison. MAKOTO achieves total reward 44.539 versus the baseline’s 41.598, a difference of +2.941 (+7.1%). Bootstrap resampling (1,000 samples) yields 95% confidence interval $[+1.947, +4.154]$, which excludes zero ($p < 0.05$).

Metric	MAKOTO	Baseline	Δ
Total Reward	44.539	41.598	+2.941 (+7.1%)
Bootstrap 95% CI	[+1.947,	+4.154]	$p < 0.05 \checkmark$
Avg. Adj. GDP Growth (%)	3.753	3.736	+0.017pp
Avg. r_growth_amplified	1.619	1.486	+0.133
Avg. Fiscal Action	+0.379	+0.332	+0.047
Avg. Monetary Action	+0.208	+0.195	+0.013
Avg. Trade Action	+0.308	+0.296	+0.011
Avg. Policy Cost	0.036	0.031	+0.005

Figure 4. MAKOTO vs. baseline agent: full performance comparison across the 24-year evaluation panel. Bootstrap CI excludes zero at 95% confidence. All reward components evaluated on held-out test trajectories.

D. Period-Level and Regime-Level Analysis

Figure 5 presents period-level policy actions across six economic epochs, with average uncertainty score ε and adjusted GDP growth for each period.

Period	Agent	ε	Fiscal	Monetary	Trade	GDP%	Δ Fiscal	
2000–07 Pre-GFC	MAKOTO	0.110	+0.370	+0.165	+0.269	+4.51%	+0.091	
	Baseline	0.110	+0.279	+0.134	+0.259	+4.48%	—	
2008–10 GFC	MAKOTO	0.264	+0.402	+0.165	+0.310	+3.17%	+0.035	
	Baseline	0.264	+0.367	+0.158	+0.334	+3.16%	—	
2011–14 Eurozone	MAKOTO	0.259	+0.474	+0.259	+0.366	+3.92%	+0.027	
	Baseline	0.259	+0.447	+0.254	+0.367	+3.92%	—	
2015–19 Normal	MAKOTO	0.315	+0.443	+0.337	+0.401	+3.44%	+0.026	
	Baseline	0.315	+0.417	+0.319	+0.363	+3.42%	—	
2020–21 COVID	MAKOTO	0.558	+0.201	+0.143	+0.204	+1.00%	−0.001	
	Baseline	0.558	+0.202	+0.168	+0.187	+1.00%	—	
2022–23 Inflation	MAKOTO	0.673	+0.198	+0.068	+0.191	+5.18%	+0.047	
	Baseline	0.673	+0.151	+0.068	+0.174	+5.17%	—	

Figure 5. Period-level policy actions and outcomes across six historical economic epochs. Δ Fiscal = MAKOTO fiscal minus baseline fiscal. The 2020–21 COVID period shows near-equivalence between agents—consistent with supply-side shock dynamics where demand stimulus has diminished efficacy.

E. Ablation Study

Figure 6 presents results from the four-condition ablation study, each trained for 100,000 timesteps under otherwise identical conditions.

Condition	Total Reward	Avg. Fiscal	Interpretation
no_cost	+47.424	+1.000	Corner solution: all instruments saturate. Confirms quadratic costs are architecturally essential.
makoto (full)	+44.539	+0.379	Full crisis conditioning + symmetric costs. Primary experimental condition.
baseline	+41.598	+0.332	No uncertainty conditioning during training. Control condition.
zero_epsilon	+41.459	+0.295	MAKOTO evaluated with $\epsilon=0$. Falls BELOW baseline: ϵ is structurally integral, not additive.

Figure 6. Ablation study across four conditions. Critical finding: zero_epsilon (41.459) falls below baseline (41.598), establishing that ϵ is a structural co-training feature rather than an additive enhancement. no_cost (fiscal = +1.000) confirms symmetric quadratic costs are necessary to prevent corner solutions.

F. Multi-Seed Robustness

Figure 7 presents robustness evaluation across three independent random seeds (100,000 timesteps per seed), confirming that the reward advantage does not depend on a single favorable initialization.

Seed	MAKOTO	Baseline	Δ Advantage
42	44.525	41.575	+2.950
7	44.516	41.576	+2.940
123	44.517	41.574	+2.943
Mean $\pm$ Std	44.519 $\pm$ 0.004	41.575 $\pm$ 0.001	+2.944 $\pm$ 0.005

Figure 7. Multi-seed robustness evaluation. MAKOTO standard deviation ± 0.004 across three seeds rules out single-seed artifact. Baseline standard deviation ± 0.001 reflects deterministic environment constraints.

IV. ANALYSIS

The results in Section III make a consistent, multi-layered case: crisis-conditioned reward architecture is not a marginal improvement to reinforcement learning-based economic policy—it is a structural prerequisite for competent performance. Four lines of evidence substantiate this claim.

The most direct evidence is the aggregate reward advantage (+7.1%, bootstrap CI [+1.947, +4.154], $p < 0.05$). This advantage does not emerge from MAKOTO accessing superior economic data—both agents observe identical 18-dimensional state vectors derived from the same World Bank panel.^[5] The difference originates exclusively in the reward architecture’s capacity to incentivize proportionally greater intervention during periods IsolationForest flags as anomalous. The amplified growth reward ($r_{\text{growth}} \times (1 + \min(\varepsilon/3, 0.4))$) encodes a macroeconomic insight with deep theoretical roots: crisis periods carry higher marginal returns to policy stimulus, consistent with Keynes’s multiplier framework,^[17] under which fiscal injection during periods of low economic activity generates above-unity output effects. Translating the 0.017 percentage point GDP growth advantage to the \$110 trillion world economy yields an estimated \$18.8 billion in annual foregone welfare under unconditioned policy—a figure that grounds an abstract reward comparison in tangible economic stakes.

The feature importance findings in Figure 2 provide independent validation that the LSTM ensemble has learned economically coherent representations rather than statistical artifacts. The three dominant predictors—lagged capital formation, gross capital formation, and lagged GFCF—directly confirm the investment accelerator mechanism formalized by Harrod:^[9] capital spending in period t generates productive capacity that drives output in $t+1$ through the multiplier-accelerator channel. That a neural network trained without any imposed economic structure independently recovers this theoretically established relationship strengthens confidence in the forecasting module’s reliability as an input to the downstream RL policy. The sixth-ranked predictor—log GDP per capita—confirms conditional convergence,^[13] and debt-to-GDP at rank seven supports the debt overhang hypothesis^[14] wherein high sovereign debt suppresses private investment through elevated risk premia. The LSTM does not merely predict GDP growth; it does so through representations that replicate half a century of growth economics.

The period-level analysis in Figure 5 reveals crisis-adaptive behavior that aggregate comparisons obscure. During 2020–21 COVID ($\varepsilon = 0.558$, the second-highest period observed), MAKOTO deploys

only +0.201 fiscal stimulus—dramatically lower than the +0.474 deployed during the 2011–14 Eurozone crisis ($\varepsilon = 0.259$). A naïve reading suggests MAKOTO provides less stimulus when uncertainty is higher, contradicting the amplification mechanism. This apparent paradox resolves once the supply-side nature of the COVID shock is considered: Guerrieri et al. establish theoretically and empirically that when production constraints rather than demand deficiency drive contraction, fiscal demand stimulus generates minimal output effects and instead accumulates as debt, triggering the quadratic penalty $r_{\text{cost}} = \text{fiscal}^2 \times \text{debt_pressure} \times 0.18$.^[18] MAKOTO’s reward architecture correctly identifies that the optimal response to a supply shock differs fundamentally from the optimal response to a demand recession—and that high ε alone does not license unlimited fiscal expansion. During 2022–23 inflation ($\varepsilon = 0.673$, the highest period), MAKOTO deploys more fiscal stimulus than the baseline (+0.198 vs. +0.151) while holding monetary action identical (+0.068 both), reflecting the stagflationary constraint where output support and price stability simultaneously compete. These period-level divergences confirm that MAKOTO has learned regime-contingent policy rules, not a uniform expansion bias.

The ablation study (Figure 6) provides causal rather than correlational evidence for the mechanism. The zero_epsilon condition—MAKOTO evaluated with ε forced to zero—achieves total reward 41.459, below the unconditioned baseline’s 41.598. This result is theoretically precise: MAKOTO was co-trained with ε as a structural feature of its reward landscape. The policy gradients shaping MAKOTO’s neural network weights were computed against a reward function that amplified growth incentives proportionally to ε . Evaluating MAKOTO without ε does not merely withhold a helpful bonus signal—it evaluates the agent in a reward environment categorically different from the one it was optimized for. The policy allocates greater stimulus in expectation of receiving amplified growth reward during high- ε periods; in a zero- ε environment, that stimulus incurs full quadratic costs without any amplified return, systematically underperforming a baseline agent that never encoded such expectations. The no_cost condition (fiscal =

+1.000, reward = +47.424) separately confirms that symmetric quadratic costs are essential: without them, the agent saturates all three policy instruments regardless of economic conditions.

The multi-seed robustness results (Figure 7; MAKOTO standard deviation ± 0.004) eliminate the single-seed artifact critique that frequently undermines RL experimental claims.^[8] All three seeds produce advantages in the range [+2.940, +2.950], with the baseline's even lower variance (± 0.001) confirming that unconditioned agents converge to a deterministic reward ceiling that the crisis-conditioned architecture consistently exceeds. Together, the bootstrap confidence interval, multi-seed replication, and causal ablation satisfy the evidentiary standards required to move from observed performance differences to mechanistic claims about reward design.

V. CONCLUSION

This paper began with a deceptively simple observation: existing RL economic policy frameworks assign identical reward weight to the 2008 Global Financial Crisis and a routine cyclical slowdown, treating one of the most consequential economic disruptions in modern history as mathematically equivalent to a 1% growth deceleration. MAKOTO resolves this design failure by embedding a continuously calibrated IsolationForest uncertainty score ε into the PPO reward function, making crisis periods structurally more consequential for the agent's learning rather than simply different in state-space coordinates.

The experimental evidence validates the falsifiable thesis: the crisis-conditioned agent achieves a statistically significant, multi-seed-robust, welfare-translatable reward advantage, and the zero_epsilon ablation establishes that this advantage reflects architectural necessity rather than parameter tuning. The broader contribution extends in three directions. Reward function design proves as consequential as neural network architecture—the most technically sophisticated agent cannot escape a reward signal that treats all periods as equivalent. IsolationForest anomaly detection, long applied to fraud and industrial

monitoring, emerges here as an underexplored source of crisis-aware signals for economic RL: it requires only a stable reference period, not labeled crisis events, making it broadly accessible. And the welfare translation—\$18.8 billion annually at current world output—connects an abstract computational experiment to the real cost of under-optimized crisis response.

MAKOTO makes no claim to being a deployable central bank policy tool. The gap between simulation and institutional implementation—encompassing political economy constraints, distributional effects, credibility mechanisms, and cross-border coordination—remains substantial. What the research does establish is a proof of concept with precise causal evidence: crisis-aware reward architecture produces measurably and structurally superior outcomes, and this superiority is a reproducible property of the design, not experimental fortune.

VI. IMPROVEMENTS

MAKOTO's results are bounded by five methodological limitations that future research must address to advance the framework toward greater validity and real-world applicability.

The most significant limitation is the IsolationForest detector's 50% recall rate against known historical crises. Six undetected events—including Germany 2009 ($\epsilon = 0.24$) and Italy 2012 ($\epsilon = 0.28$)—are characterized by policy interventions that suppressed standard crisis indicators despite severe economic disruptions. Germany's Kurzarbeit short-time work programme preserved headline unemployment metrics even as GDP contracted 5.7%, rendering the crisis invisible to a detector trained on standard macroeconomic variables. Three targeted improvements would address this: expanding the crisis feature set to include financial market indicators (sovereign credit default swap spreads, equity volatility indices, exchange rate pressure indices) that capture banking-sector stress independently of real-economy outcomes; replacing the binary IsolationForest threshold with a variational autoencoder that learns a continuous latent representation of economic regime, enabling smoother uncertainty gradations;

and implementing an ensemble of anomaly detectors trained on multiple reference sub-periods to reduce sensitivity to the specific calibration window.

The second limitation is annual data resolution. World Bank indicators are published annually, but crisis dynamics—particularly financial contagion and monetary transmission—evolve over weeks to months. Integration of higher-frequency data (IMF International Financial Statistics at quarterly resolution, BIS credit data) would enable the LSTM ensemble to capture crisis onset with greater temporal fidelity, potentially improving both forecast accuracy and crisis detection recall above the current 50% threshold.

Third, the policy transmission coefficients (fiscal multiplier = 0.30, monetary = 0.15, trade = 0.10) are calibrated by the researcher rather than empirically estimated. Future implementations should derive these parameters through panel regression on the World Bank dataset or draw from published meta-analyses of fiscal multipliers across economic regimes, eliminating this degree of calibration freedom and strengthening the claim of policy realism.

Fourth, MAKOTO's single-agent design abstracts away the game-theoretic interactions between national policymakers that dominate real-world economic coordination. A multi-agent extension—where each country operates its own PPO agent and MAKOTO's coordination signal serves as a shared communication channel—would more realistically model strategic interdependence and could reveal whether crisis-conditioned reward design generates coordination benefits beyond the single-planner optimum demonstrated here.

Finally, the 30-country panel underrepresents 165 United Nations member states. The framework's welfare implications are computed at world GDP scale but calibrated on a high-income-biased sample. Extending coverage to low-income countries—requiring specialized imputation strategies for substantially higher rates of World Bank data missingness—would strengthen the generalizability of findings across the full income distribution.

VII. EVALUATION

Evaluating MAKOTO as a research endeavor requires honest assessment across three dimensions: methodological validity, goal achievement, and insights generated through the research process itself.

On methodological validity, the research design’s strongest features are its strict temporal data separation—training, validation, and test sets are non-overlapping chronological windows that prevent information leakage—and the multi-seed robustness evaluation that guards against false positives arising from favorable random initialization, a pervasive validity threat in RL experimental literature.^[8] The bootstrap confidence interval provides statistically rigorous quantification of the reward advantage, and the four-condition ablation design directly tests the causal mechanism rather than relying on correlation between ϵ and performance. These design choices collectively satisfy the minimum evidentiary standards for RL experimental claims. The most significant methodological weakness is the absence of an empirically calibrated policy transmission model—the researcher-set multipliers (fiscal: 0.30, monetary: 0.15, trade: 0.10) introduce a degree of simulation realism uncertainty that cannot be fully addressed without structural econometric estimation.

On goal achievement, MAKOTO accomplished its primary objective: demonstrating through controlled experiment and causal ablation that crisis-conditioned reward architecture produces a statistically significant and structurally necessary reward advantage over an unconditioned baseline. The 7.1% advantage with bootstrap confidence interval excluding zero at 95% constitutes a stronger result than initially projected; pre-experimental expectations anticipated a significant but modest effect in the 2–4% range. The multi-seed robustness (± 0.004 standard deviation) exceeded the target robustness threshold. The research did not fully achieve the secondary objective of high crisis detection recall: the 50% recall rate falls short of the 75–80% threshold at which crisis signals would provide reliable enough coverage for real-world deployment consideration.

The most intellectually significant outcome of the research process was unanticipated. The zero_epsilon ablation finding—that MAKOTO without its crisis signal underperforms the unconditioned baseline—was not part of the original experimental design. It reframes the contribution from ‘crisis conditioning helps’ to ‘crisis conditioning is a structural requirement,’ a qualitatively stronger claim that carries direct implications for reward function design across any RL domain exhibiting environmental regime heterogeneity. The same architectural principle—embedding a real-time regime signal as a structural reward multiplier rather than an additive bonus—may generalize to healthcare resource allocation under surge conditions, climate policy optimization under tipping-point uncertainty, and emergency logistics under disaster scenarios. Identifying this generalization potential, unanticipated at the outset, represents the primary intellectual contribution of the research journey beyond the specific MAKOTO results.

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